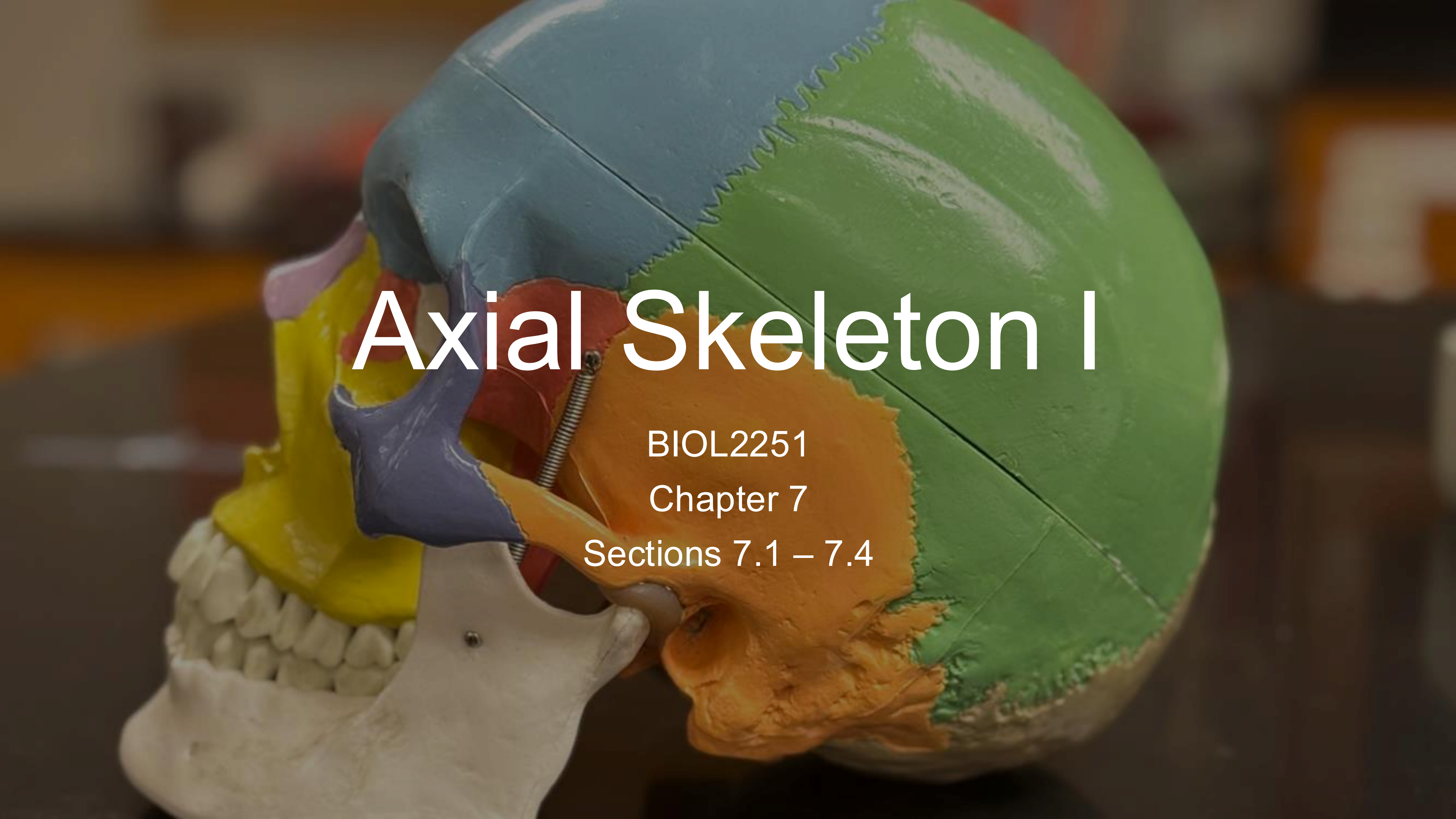


Axial Skeleton I

BIOL2251

Chapter 7

Sections 7.1 – 7.4

Axial vs. Appendicular Sort

Axial vs. Appendicular Sort

- **Instructions**
  - Work on your own
  - I will project a list of bones
    - **Sort bones by axial skeleton and appendicular skeleton**
  - We will review as a class
- **Bloom’s taxonomy = Remember**

Axial vs. Appendicular Sort

- Skull
- Clavicle
- Sternum
- Femur
- Ribs
- Humerus
- Vertebral column
- Radius
- Sacrum
- Patella
- Hyoid
- Ulna
- Os coxae (hip bones)
- Fibula
- Auditory ossicles
- Scapula

Axial vs. Appendicular Sort

Axial

- Skull
- Sternum
- Ribs
- Vertebral column
- Sacrum
- Hyoid
- Auditory ossicles

Appendicular

- Clavicle
- Femur
- Humerus
- Radius
- Ulna
- Patella
- Os coxae/hip bones
- Fibula
- Scapula

Name That Bone

Name That Bone

- **Instructions**
  - Work on your own
  - I will project a table giving descriptions of different bones of the axial skeleton
    - **Based on the definition, identify the bone**
  - Compare your answers with a partner
- **Bloom's taxonomy = Remember, Understand**

Name That Bone

Description	Answer
Single bone forming the forehead and the superior wall of each orbit	Frontal
Two bones forming the superior and lateral walls of the cranium; meet at the sagittal suture	Parietal (×2)
Two bones forming the lateral skull inferior to the parietal bones; house the middle and inner ear	Temporal (×2)
Single bone forming the posterior skull and base; contains the foramen magnum	Occipital
Butterfly-shaped bone forming the central floor of the cranium; visible on lateral skull	Sphenoid
Delicate bone forming the roof of the nasal cavity and part of the medial orbital wall; contains the cribriform plate	Ethmoid

Bone Structure Competition

Bone Structure Competition

- **Instructions**
  - Work in small groups (12 total)
  - I will read a description of a bone structure
    - **In your groups, you will identify the structure and write it on your mini-whiteboard**
      - **You will have 30 seconds per structure**
  - Correct structure identification = 1 point
  - Team with the most points wins!
- **Bloom’s taxonomy = Remember, Understand**

Bone Structure Competition

Clue Read Aloud	Answer
Large opening through which the spinal cord connects to the brain.	Foramen magnum → Occipital
Two rounded projections that articulate with the first cervical vertebra.	Occipital condyles → Occipital
Bony ridge you feel just above your eye socket that helps to protect the eye	Supra-orbital margin → Frontal
A projection of the temporal bone that forms part of the zygomatic arch.	Zygomatic process → Temporal
A rough, rounded bump posterior to the ear; insertion point for neck muscles that rotate the head or flex the neck.	Mastoid process → Temporal
A thin, spike-like projection inferior to the ear; anchors ligaments and muscles of the tongue and throat.	Styloid process → Temporal
A canal running through the temporal bone that carries the internal carotid artery.	Carotid canal → Temporal
Opening through which vestibulocochlear nerve (CN VIII) enters the ear; ends at the eardrum.	External acoustic meatus → Temporal
A foramen through which multiple cranial nerves (CN IX, X, XI) pass, alongside the internal jugular vein.	Jugular foramen → Temporal/Occipital border
Twin openings through which the optic nerves pass from the eyes to the brain.	Optic canals → Sphenoid
Saddle that cradles the pituitary gland.	Sella turcica → Sphenoid
Forms the roof of the nasal cavity and a connection point for the cranial dura mater.	Cribriform plate → Ethmoid

Orbital and Nasal Complex Build

Orbital and Nasal Complex Build

- **Instructions**
  - Work in pairs
  - **Without looking at your notes, list all the bones in the:**
    - **(A) Orbital complex**
    - **(B) Nasal complex**
  - After writing your list, check against your notes
  - We will review as a class
- **Bloom’s taxonomy = Understand, Analyze**

Orbital and Nasal Complex Build

Orbital Complex

- Frontal (roof/superior wall)
- Sphenoid (posterior wall/lateral)
- Zygomatic (lateral wall/floor)
- Maxilla (floor/medial floor)
- Palatine (floor*)
- Lacrimal (medial wall)
- Ethmoid (medial wall)

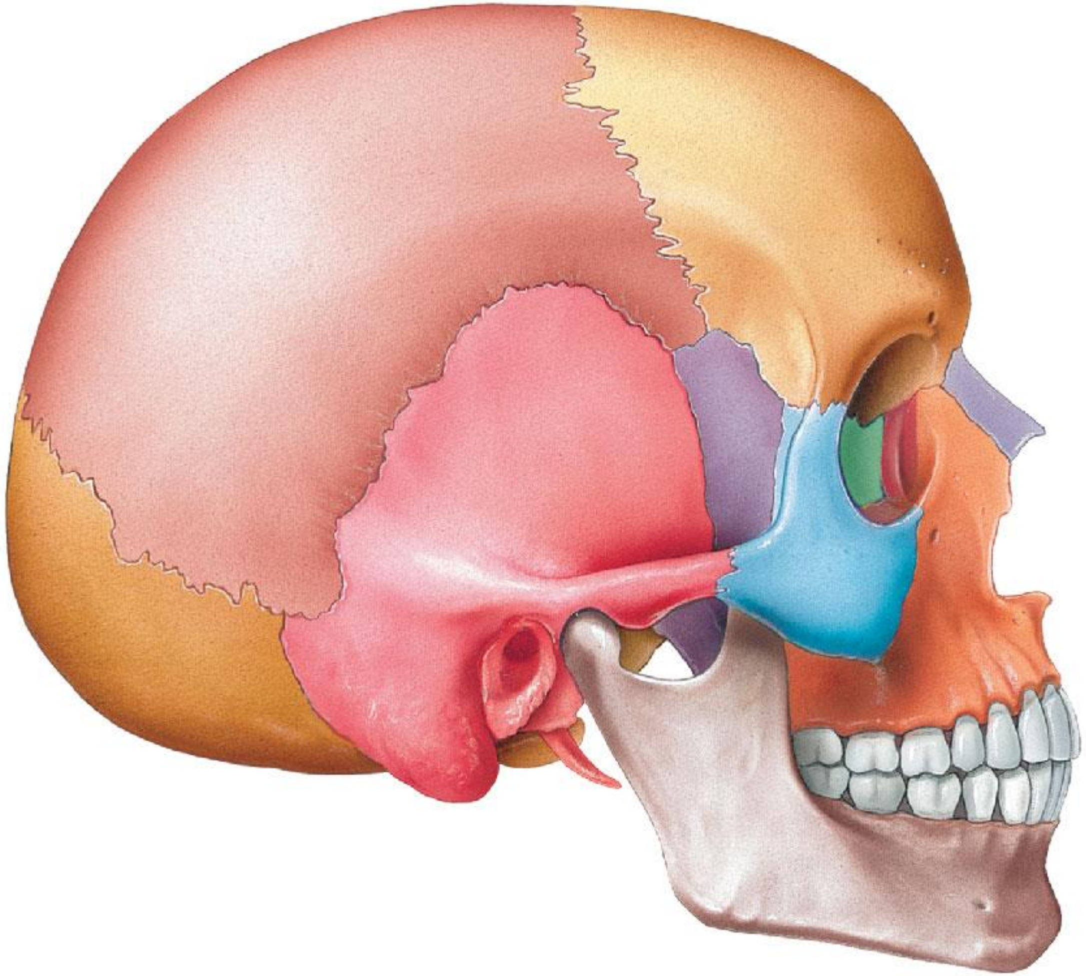
Nasal Complex

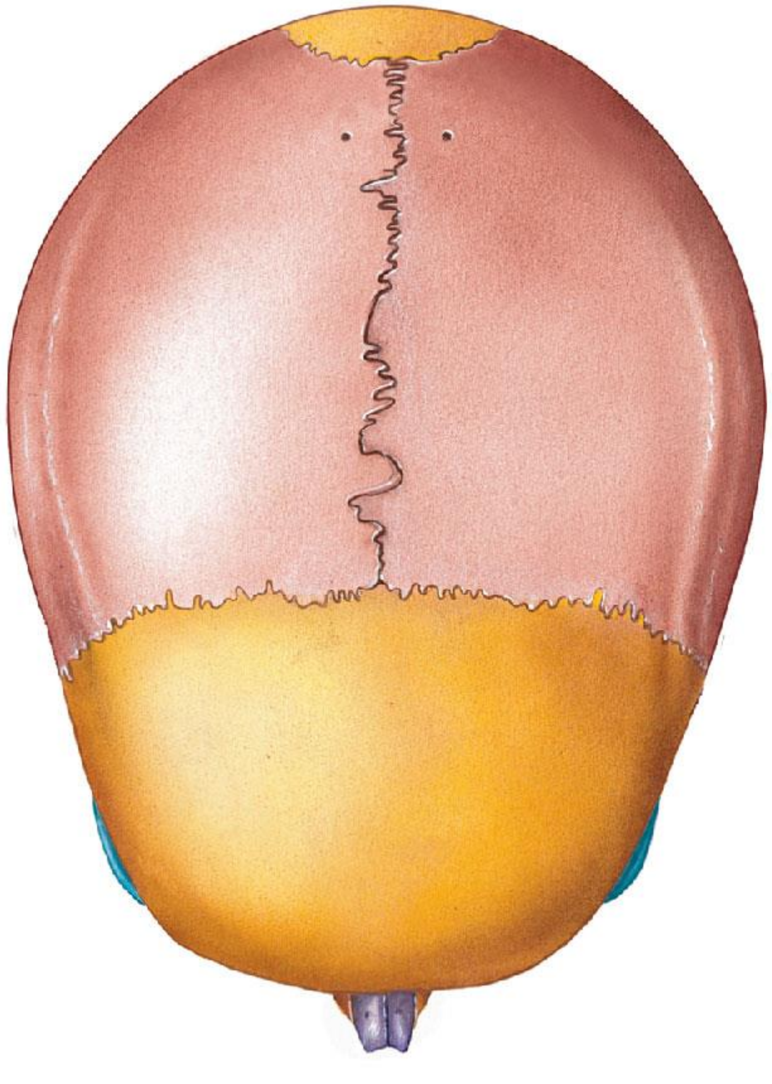
- Nasal bones (bridge)
- Maxilla (lateral nasal walls/floor)
- Palatine (posterior floor)
- Ethmoid (lateral walls + roof)
- Inferior nasal concha (lateral walls)
- Vomer (inferior nasal septum)
- Sphenoid (roof/posterior)

Suture Mapping

Suture Mapping

- **Instructions**
  - Work in small groups (12 total)
  - Sketch a simplified superior view and lateral view of the skull
  - **Draw and the label the four main sutures**
    - Coronal
    - Squamous
    - Sagittal
    - Lambdoid
  - **Indicate which bones are connected by each suture**
  - Group(s) will volunteer to share their labels and bone connections
- **Bloom’s taxonomy = Remember, Understand**





Suture Mapping

Suture	Bones Connected	View Best Seen
Sagittal	Left parietal + Right parietal	Superior
Coronal	Frontal + Parietal (both)	Superior / Lateral
Lambdoid	Occipital + Parietal (both)	Posterior / Superior
Squamous	Temporal + Parietal (each side)	Lateral

Fontanelle Mini-Case

Fontanelle Mini-Case

- **Instructions**
  - **Think on your own about the following clinical case**
    - Answer each of the questions, then discuss with a partner
  - Pairs will share with a neighboring pair
    - Final review as a whole class
- **Bloom's taxonomy = Understand, Apply**

Fontanelle Mini-Case

A new mother brings her 3-month-old infant to a well-child visit. During assessment, the nurse notes that the anterior fontanelle feels tense and bulging. The infant is fussy and has had a fever for two days.

1. What is a fontanelle, and what is it made of?
2. Where is the anterior fontanelle located, and which bones border it?
3. What does a bulging fontanelle suggest, and why?

Fontanelle Mini-Case

- **Q1:** A fontanelle is a fibrous gap between cranial bones in the infant skull. The membrane allows for compression during birth and then rapid brain growth
- **Q2:** Anterior fontanelle — located at the junction of the sagittal and coronal sutures; bordered by the two frontal and two parietal bones. It is the largest of the fontanelles.
- **Q3:** A bulging, tense fontanelle suggests increased intracranial pressure, which in an infant raises concern for meningitis or encephalitis. A sunken fontanelle would suggest dehydration.

Who Am I?
Bone Riddles

Who Am I? Bone Riddles

- **Instructions**
  - Work on your own
  - I have a list of riddles to read aloud
    - **Each riddle indicates a bone, structure, suture, or fontanelle for you to identify**
  - First person with their hand up gets the opportunity to share
  - Correct identification = small prize
- **Bloom’s taxonomy = Analyze**

Who Am I? Bone Riddles

- I form your forehead and the superior roof of each eye socket. Just above each orbit I have a palpable ridge that you can feel on yourself right now. I am unpaired and sit at the very front of the cranium. Who am I?

→ **Frontal Bone**

- I am the largest foramen in the skull. The spinal cord, vertebral arteries, and accessory nerve all pass through me. I sit on the inferior surface of the occipital bone. Without me, the brain and body could not communicate. Who am I?

→ **Foramen Magnum**

Who Am I? Bone Riddles

- I am actually two bones fused at the midline, though I function as one. I form your upper jaw and anchor your upper teeth, contribute to the hard palate, the floor of the nasal cavity, and most of the orbital floor. I am the most central bone of the face. Who am I?

→ **Maxilla**

- I am the bony margin that forms the complete border around the eye socket. I am not a single bone — the frontal bone forms my top, the zygomatic bone my lateral portion, and the maxilla my inferior and medial portions. Together we form a protective ring around the eye. Who am I?

→ **Orbital Rim (Frontal, Zygomatic, and Maxilla)**

Who Am I? Bone Riddles

- I run along the midline of the skull from front to back, connecting the two parietal bones to each other. My name refers to an arrow, reflecting my straight anterior-to-posterior course. I am best seen from a superior view of the skull. Who am I?

→ **Sagittal Suture**

- I am the largest fontanelle and the one nurses and clinicians assess most often. I sit at the junction of the sagittal and coronal sutures, bordered by the two frontal and two parietal bones. I am the last fontanelle to close, remaining open until 18–24 months after birth. If I am enlarged or bulging, this suggests increased intracranial pressure. Who am I?

→ **Anterior Fontanelle**